

Ehsan Ghoreishi

Virginia Tech, Blacksburg, VA, U.S.

✉ ehsangh@vt.edu **in** LinkedIn  Google Scholar

EDUCATION

Virginia Tech Ph.D. in Computer Engineering	Blacksburg, VA, U.S. 2024-present
Virginia Tech M.Sc. in Computer Engineering	Blacksburg, VA, U.S. 2024
Isfahan University of Technology B.Sc. in Electrical Engineering	Isfahan, Iran 2021

RESEARCH EXPERIENCE

Real-Time GPU Acceleration for URLLC/eMBB Multiplexing Complex Network & Security Research, Virginia Tech	2022-present
Deep Reinforcement Learning Solution for URLLC/eMBB Multiplexing Complex Network & Security Research, Virginia Tech	2022-2024
Complex-Valued Neural Network for Wireless Modulation Classification Isfahan University of Technology Signal Processing Research Center	2020-2021
Designed and Analyzed Nonlinear Effects in Remote Wire Monitoring. Isfahan University of Technology Optic Research Center	2020-2021
Calculation of Heat Distribution of Double Brillouin Pulse in Water Isfahan University of Technology	2021

WORK EXPERIENCES

Graduate Research Assistant, CNSR Lab, Virginia Tech, Blacksburg, VA
Graduate Teaching Assistant, Bradley Dept. of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA
Research Assistant, Electrical and Computer Department, Isfahan University of Technology, Isfahan, Iran.
Research Assistant, Optical and Signal Processing Research Center, Isfahan University of Technology, Isfahan, Iran.

PUBLICATIONS

Papers.....
Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN." in 2024 33rd International Conference on Computer Communications and Networks (ICCCN), 2024, pp. 1–9. (Invited Paper)
Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus+: A Real-Time DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN.", (Submitted to *IEEE Transactions on Machine Learning in Communications and Networking*)
Ehsan Ghoreishi, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Enhancing 5G Efficiency: How Machine Learning is Revolutionizing eMBB and URLLC Puncturing.", (To be submitted to *IEEE Wireless Communications*)

Poster.....
Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN." Office of Naval Research (ONR) MURI Meeting, 2024.

AWARDS AND ACHIEVEMENTS

CCI SWVA Cyber Innovation Scholar Recipient, Commonwealth Cyber Initiative – SWVA	2025
Nationwide University Entrance Exam – Mathematical Physics Ranked 90 th among 200,000 participants nationwide	2017
Ranked 15th in cumulative GPA in ECE Department of Isfahan University of Technology Among 185 Electrical Engineering students	2020

TEACHING EXPERIENCE

The Graduate Teaching Assistant in Introduction to Computer Networking Bradley Dept. of Electrical and Computer Engineering, Virginia Tech	2022
The Teaching Assistant in Principles of Electronics Electrical and Computer Engineering Dept., Isfahan University of Technology	2021
The Teaching Assistant in Electric Circuits II Electrical and Computer Engineering Dept., Isfahan University of Technology	2021
Course Instructor in MATLAB Electrical and Computer Engineering Dept., Isfahan University of Technology	2022

SKILLS

Programming: Python, MATLAB, C/C++, Assembly
Experienced with numerical and simulation tools: TensorFlow, TF-Agents, PyTorch, MATLAB, Anaconda, CST, Maxwell, MATLAB Simulink

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE)	2022-present
IEEE Communications Society (ComSoc)	2022-present

RESEARCH INTERESTS

Machine Learning
Reinforcement Learning
Real-Time Systems
Optimization
Wireless Networks